

SPTECHPIONEER

COMPLETE ERP WORKFLOW DOCUMENT (DETAILED & IMPLEMENTATION-READY)

1. INTRODUCTION

This document is the **official internal ERP process document of SPTECHPIONEER**. It defines **how SPTECHPIONEER operates day-to-day**, how data flows between departments, and what rules **must be followed inside the company**.

It is written **specifically for SPTECHPIONEER's manufacturing, machining, subcontracting, and project-based execution model**.

This document shall be treated as:

- Internal Company Process Manual
- ERP Usage Rulebook
- Department Responsibility Charter
- Audit & Management Reference

Any deviation from this document must be **formally approved by Management**.

2. SPTECHPIONEER – CORE OPERATING PRINCIPLES

These principles define **how SPTECHPIONEER runs its business inside the ERP**. Every department must strictly follow these rules.

2.1 Department Ownership Principle

At SPTECHPIONEER, **each department owns its data**. No department is allowed to modify data belonging to another department.

Department Ownership Principle

- Sales → Customer commitment data
 - Design → Engineering & BOM data
 - Purchase → Supplier & rate data
 - Store → Physical quantity movement
 - Quality → Acceptance/Rejection decisions
 - Production → Execution status
 - Accounts → Financial posting
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- Sales owns customer commitment data
 - Design owns engineering and BOM data
 - Purchase owns supplier and rate data
 - Stores owns physical quantity movement
 - Quality owns acceptance and rejection decisions
 - Production owns execution status
 - Accounts owns financial posting

This rule prevents confusion, blame shifting, and data conflicts.

2.2 Forward-Only Process Principle

All business processes at SPTECHPIONEER follow a **forward-only flow**.

Once a document is approved:

- It cannot be edited backward
- Any correction requires a controlled revision or new document

This ensures traceability and audit compliance.

2.3 Engineering-Driven Costing Principle

At SPTECHPIONEER:

Cost is not guessed. Cost is engineered.

All costing originates from:

- Approved BOM
- Approved routing
- Approved operation rates

Sales, Purchase, and Accounts **consume cost data** but do not define it.

2.4 Quality Gate Principle

No material or job can move forward in SPTECHPIONEER without Quality approval.

Quality decisions override:

- Production urgency
- Purchase pressure
- Delivery pressure

This ensures customer satisfaction and repeat business.

2.5 Project-Centric Profit Principle

SPTECHPIONEER measures profitability **per job / project**, not only per invoice.

All costs and revenues must be linked to:

- Job Code
- Project Code

Management decisions are taken based on **project-level P&L**, not assumptions.

3. DEPARTMENT-WISE WORKFLOW

3.1 ADMIN DEPARTMENT

(System Governance & Control)

Purpose: Admin establishes and controls the ERP environment. Admin ensures system stability, access control, and audit compliance.

Responsibilities

- Company Master creation
- Financial Year definition
- User & Role management
- Approval matrix setup
- Document numbering rules
- Audit trail configuration

Key Data Managed

- Company Name, GST, Address
- Financial Year
- User Roles & Permissions
- Approval Hierarchies
- Audit Logs

System Rules

- Admin does not create operational transactions
- Admin actions are logged permanently

Output

- Fully configured ERP system

3.2 SALES DEPARTMENT

(Customer Commitment Owner)

Purpose: Sales converts customer demand into executable internal jobs and provides formal pricing through quotations.

Workflow

1. Customer Enquiry / RFQ received
2. **Quotation created**
3. Quotation approved (Sales Head / Management)
4. Customer PO received (based on approved quotation)
5. PO and Drawing uploaded
6. Sales Order created
7. Sales Order approved
8. Job / Project Code auto-generated

Data Ownership

- Customer Name
- Customer PO Number
- PO Date
- Delivery Date
- Item & Quantity
- Drawing Number & Revision

- Drawing PDF
- Overhead %
- Profit %
- Quotation Number / Quotation Details

System Behavior

- Duplicate Quotation / PO validation
- Data locked after approval
- Job code becomes primary reference

Restrictions

- No BOM editing
- No stock access
- No invoice posting

Output

- Approved Quotation
- Approved Sales Order
- Job / Project Code

Workflow Diagram

Customer Enquiry / RFQ



Quotation Creation



Quotation Approval



Customer PO Received



PO & Drawing Upload



Sales Order Creation



Sales Order Approval



Job / Project Code Generated

Sidebar / Points

- Customer Name, PO Number, PO Date
 - Delivery Date, Item & Quantity
 - Drawing Number & Revision
 - Drawing PDF Upload
 - Overhead %, Profit %
 - Quotation Number / Details
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- Restrictions: No BOM Editing, No Stock Access, No Invoice Posting

3.3 DESIGN DEPARTMENT

(Engineering Authority)

Purpose: Design translates customer requirements into manufacturable specifications.

Workflow

1. Receives approved Sales Order
2. Uploads controlled drawing
3. Creates BOM
4. Defines routing & operations
5. Submits BOM for approval

Data Ownership

- Drawing Number & Revision Control
- Raw Material BOM
- Bought-out Items
- Operation List (VMC, CNC, Drilling, Inspection, Packing)
- Process Sheet
- Inspection Parameters

System Behavior

- BOM version control
- BOM lock after approval
- Cost freeze after approval

Output

- Approved BOM
- Approved Routing

Workflow Diagram:

Receive Approved Sales Order



Upload Controlled Drawing



Create BOM & Routing



Define Operations & Process Sheet



BOM Approval & Cost Freeze

Sidebar / Points:

- Drawing Number & Revision Control
- Raw Material & Bought-Out Items
- Operation List (VMC, CNC, Drilling, Inspection, Packing)
- Process Sheet & Inspection Standards

3.4 PURCHASE DEPARTMENT

(External Procurement Controller)

Purpose: Purchase ensures timely and cost-effective procurement.

Workflow

1. System-generated purchase requirement from BOM

2. Supplier selection
3. Purchase Order creation
4. Purchase Order approval

Data Ownership

- Supplier
- PO Number
- Item
- Quantity
- Rate
- Delivery Date
- Drawing Reference
- Subcontract Charges

System Behavior

- Rate visibility restricted
- PO sent to Store for GRN reference

Restrictions

- No GRN creation
- No invoice posting

Output

- Approved Purchase Order

Workflow Diagram:

Generate Purchase Requirement



Supplier Selection



Purchase Order Creation



Purchase Order Approval



P0 Sent to Store for GRN Reference

Sidebar / Points:

- Supplier, Item, Quantity, Rate
- Delivery Date & Drawing Reference
- Subcontract Charges
- Restrictions: No GRN Creation, No Invoice Posting

3.5 STORE DEPARTMENT

(Physical Inventory Controller)

Purpose: Store ensures physical stock accuracy and traceability.

Workflow

1. Material receipt against PO
2. GRN creation

3. Warehouse allocation
4. Material issue to production

Data Ownership

- GRN Number
- Item
- Received Quantity
- Warehouse Type (RM / WIP / FG / Subcontract)
- Issue Quantity
- Transfer Quantity

System Behavior

- Stock Ledger auto-update
- Stock Balance auto-update
- Rate hidden from Store

Output

- Updated stock records

Workflow Diagram:

Material Receipt against PO



GRN Creation



Warehouse Allocation



Material Issue to Production / WIP



Stock Ledger & Balance Update

Sidebar / Points:

- GRN Number, Item, Received Quantity
- Warehouse Type (RM / WIP / FG / Subcontract)
- Issue & Transfer Quantity
- Restrictions: Rate & Value Hide

3.6 QUALITY DEPARTMENT

(Process Gatekeeper)

Purpose: Quality ensures defective material never flows forward.

Workflow

- Incoming QC after GRN
- In-process QC during production
- Final QC before FG storage

Data Ownership

- Inspection Type
- Accepted Quantity
- Rejected Quantity
- Rework Quantity
- QC Status
- Inspection Report Upload

System Behavior

- Rejected material blocked
- Only QC-approved material moves forward

Output

- QC Approval / Rejection

Workflow Diagram:

Incoming QC (After GRN)



In-Process QC (During Production)



Final QC (Before FG Storage)



QC Approval / Rejection

Sidebar / Points:

- Inspection Type, Accepted / Rejected / Rework Qty
- QC Status & Inspection Report Upload
- Controls Next Process Flow

3.7 VMC / CNC PROCESSING DEPARTMENT

(Machine Performance Tracking)

Purpose: Tracks machine utilization and machining cost.

Workflow

- Machine operation logging

- Production quantity update
- Rejection recording

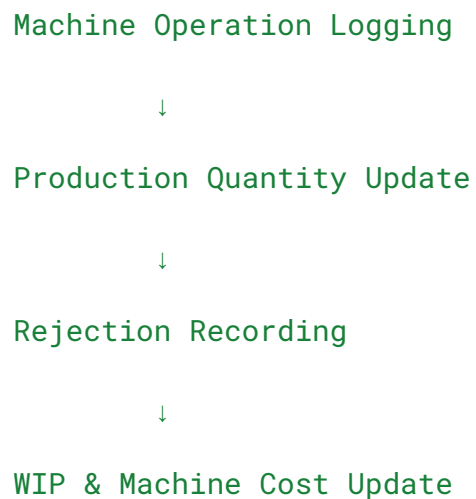
Data Ownership

- Machine Name
- Program Number
- Cycle Time
- Runtime
- Produced Quantity
- Rejection Quantity

System Behavior

- Machine cost auto-calculated
- WIP updated

Workflow Diagram:



Sidebar / Points:

- Machine Name, Program Number, Cycle Time, Runtime
- Produced & Rejected Quantity
- Cost Auto-Calculation

3.8 PRODUCTION DEPARTMENT

(Execution Controller)

Purpose: Ensures job completion as per plan.

Workflow

1. Work Order creation
2. Routing execution
3. WIP tracking
4. Job completion
5. Job closure

Data Ownership

- Work Order Number
- Job Code
- Routing Sequence
- Target Quantity
- WIP Status
- Completion Status

Status Flow: Draft → In Progress → Completed → Closed

System Behavior

- Cannot close without QC clearance

Workflow Diagram:

Work Order Creation



Routing Execution



WIP Tracking



Job Completion



Job Closure (QC Cleared)

Sidebar / Points:

- Work Order Number, Job Code, Routing Sequence
- Target Quantity, WIP Status, Completion Status
- Status Flow: Draft → In Progress → Completed → Closed

3.9 ACCOUNTS DEPARTMENT

(Financial Authority)

Purpose: Ensures statutory compliance and profitability analysis.

Workflow

- Purchase invoice booking

- Sales invoice generation
- GST posting
- Payment & receipt tracking

Data Ownership

- Purchase Invoice
- Sales Invoice
- GST %
- Tax Amount
- Ledger Entries

System Behavior

- No stock access

Output

- P&L
- Invoice-wise profit
- Project-wise profit

3.10 AUTOMATION PROJECT DEPARTMENT

(Project-Based Execution)

Purpose: Handles non-standard or automation projects.

Workflow

- Project creation
- Task assignment

- Timesheet entry
- Expense tracking
- Revenue mapping

Output

- Project-wise P&L

Workflow Diagram:

Project Creation



Task Assignment



Timesheet Entry



Expense Tracking



Revenue Mapping



Project P&L

Sidebar / Points:

- Project Code, Task List
- Timesheet & Expense Entry
- Material Cost & Revenue
- Output: Project-wise Profitability

SPTECHPIONEER ERP – MAIN CONCERNS / KEY POINTS (Point-wise)

1. ADMIN DEPARTMENT

- Admin sets up ERP **company master, financial year, user roles, approval matrix, and audit logs.**
- **Main Concern:** Admin **does not touch operational data**; ensures **system stability and control.**

2. SALES DEPARTMENT

- Responsible for **customer commitment and order execution.**
- **Key Concerns:**
 1. **Quotation Creation:** Sales creates quotation **before customer PO.**
 2. Quotation must be **approved before PO** → ensures pricing and profit control.
 3. Sales Order is created **only after customer confirms Quotation.**
 4. Job / Project Code is auto-generated from approved Sales Order.
 5. Sales **cannot edit BOM, access stock, or post invoices.**
 6. **Data traceability:** Customer → Quotation → PO → Sales Order → Job Code.

3. DESIGN DEPARTMENT

- Responsible for **engineering and manufacturable specifications.**
- **Key Concerns:**
 1. Designs are created **only after approved Sales Order.**
 2. BOM, Routing, and Process Sheet must be **approved and locked.**
 3. Cost is **engineered and frozen**; Sales/Accounts cannot modify.
 4. Revision control is mandatory for drawings and BOM.

4. PROCUREMENT / PURCHASE DEPARTMENT

- Handles **approved purchase requirements** from BOM.
- **Key Concerns:**
 1. Procurement acts only on **approved BOM and Sales Orders**.
 2. Supplier selection, PO creation, and PO approval **must follow ERP rules**.
 3. Procurement cannot create GRN or post invoices.
 4. **Rate visibility is restricted**; only approved rates are used.
 5. Ensures **timely material procurement** without bypassing controls.

5. STORE DEPARTMENT

- Manages **physical stock movement**.
- **Key Concerns:**
 1. GRN created **only against approved PO**.
 2. Warehouse allocation, material issue, stock ledger, and stock balance are auto-updated.
 3. Rate/value is **hidden**; quantity only visible.
 4. Forward-only material flow; cannot modify incoming/outgoing quantities arbitrarily.

6. QUALITY DEPARTMENT

- Acts as **process gatekeeper**.
- **Key Concerns:**
 1. Material cannot move forward without **QC approval** (Incoming, In-process, Final).
 2. Rejected material is **blocked or returned**.
 3. Ensures **customer satisfaction and defect-free output**.
 4. QC decisions **override production urgency**.

7. VMC / CNC PROCESSING DEPARTMENT

- Tracks **machine operations and WIP generation**.
- **Key Concerns:**
 1. Production data logged with **machine runtime, cycle time, program number**.
 2. WIP quantity and rejection automatically updated.
 3. Machine cost is **calculated automatically**; no manual intervention.

8. PRODUCTION DEPARTMENT

- Manages **execution and job completion**.
- **Key Concerns:**
 1. Work Order execution **depends on BOM and QC-approved materials**.
 2. WIP tracking must be accurate.
 3. Job cannot close **without Final QC approval**.
 4. Status flow: Draft → In Progress → Completed → Closed.

9. ACCOUNTS DEPARTMENT

- Handles **financial posting and statutory compliance**.
- **Key Concerns:**
 1. Accounts **never touch stock quantities**.
 2. Purchase and Sales invoices posted **only after approved documents**.
 3. Profit analysis done per **invoice and project**.
 4. GST and ledger entries must match ERP-controlled documents.

10. AUTOMATION / PROJECT DEPARTMENT

- Manages **special or automation projects**.

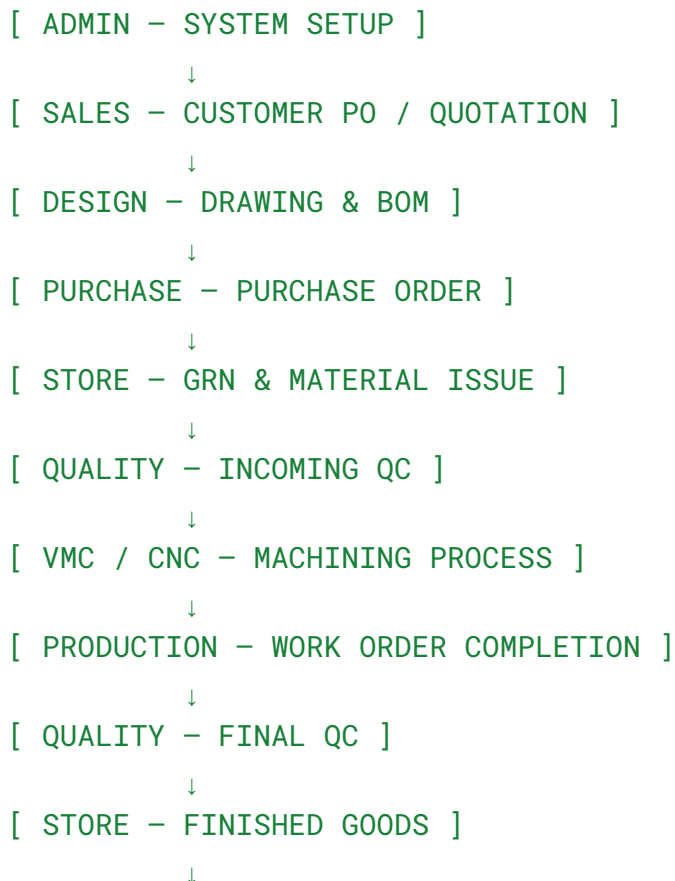
- **Key Concerns:**

1. Project creation and task assignment must be **linked to Job / Project Code**.
2. Timesheets, expenses, and revenue mapped **per project**.
3. Project-wise P&L generated **only after all approvals**.

4. END-TO-END FLOW SUMMARY

Admin Setup → Sales Order → Design BOM → Purchase Order → Store GRN → Quality Inspection → Production Execution → Final QC → Finished Goods → Accounts Invoicing → Project Profit Analysis

5. SPTECHPIONEER – END-TO-END VISUAL FLOW (CLIENT VIEW)



[ACCOUNTS – INVOICE & PROFIT]



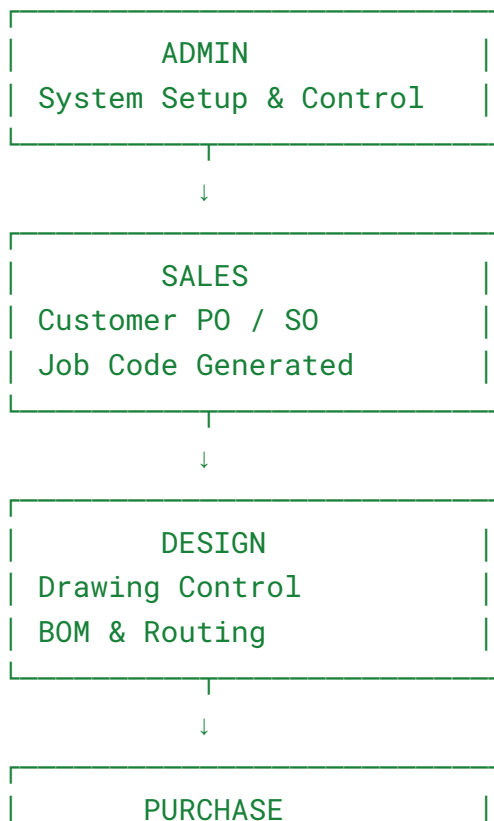
[AUTOMATION PROJECT – PROJECT P&L (IF APPLICABLE)]

6. MANAGEMENT CONTROL SUMMARY

- No department overlaps another department's responsibility
- Every movement is approval-driven
- Quality controls forward flow
- Engineering controls cost
- Projects capture profitability

7. SPTECHPIONEER – WORKFLOW REPRESENTATION DIAGRAMS

7.1 MASTER END-TO-END ERP WORKFLOW DIAGRAM (SYSTEM VIEW)



Purchase Order
Subcontract PO



STORE
GRN Creation
Stock Ledger Update



QUALITY
Incoming QC
Accept / Reject



VMC / CNC PROCESS
Machining Execution
WIP Generation



PRODUCTION
Work Order Completion
Job Closure



QUALITY
Final QC



STORE
Finished Goods Stock



ACCOUNTS
Invoice & GST

Project P&L	
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7.2 DEPARTMENT-WISE SWIMLANE WORKFLOW (RESPONSIBILITY VIEW)

ADMIN

Company FY Users Approvals

SALES

Customer PO → Sales Order → Job Code

DESIGN

Drawing → BOM → Routing → Cost Freeze

PURCHASE

Supplier Selection → PO Approval

STORE

GRN → Stock Ledger → Material Issue

QUALITY

Incoming QC → In-Process QC → Final QC
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VMC / CNC

Machining → Production Qty → Rejection
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PRODUCTION

Work Order → Execution → Completion

STORE

FG Stock Entry

ACCOUNTS

Invoice → GST → Project Profit

7.3 JOB / PROJECT LIFE CYCLE FLOW (JOB-CENTRIC VIEW)

Customer PO



Sales Order



Job Code Created



BOM & Routing Approved



Material Procured



Material Received (GRN)



QC Approved



Production Started



Machining Completed



Final QC Approved



Finished Goods



Invoice Generated



Job Closed



Project P&L Calculated

Material / Job



FINANCIAL FLOW

Purchase Invoice

GST Posting

Expense Booking

Sales Invoice

Profit Analysis

- ## 7.6 FINAL ERP CONTROL SUMMARY DIAGRAM (MANAGEMENT VIEW)

QUALITY → Controls Flow

STORE → Controls Quantity

ACCOUNTS → Controls Money

PROCUREMENT → Controls Approved Purchases

MANAGEMENT → Controls Pol

8. FINAL END-TO-END ERP WORKFLOW STATEMENT (OFFICIAL)

At SPTECHPIONEER, every job follows a **controlled, forward-only, quality-gated, engineering-driven ERP workflow**.

This ERP workflow is **mandatory, auditable, and management-approved**.